

TIC 307453359

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.740101 d
Epoch (t0) 1493.4111 BTJD
Duration 0.0840 d
Depth 7.111 % (71,106 ppm)
SNR 107.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 6 (gate ≥ 3)
Robust depth ratio 1.05 (gate ≥ 0.50)
Robust depth sigma 89.0 (gate ≥ 7.0)
Odd/even z-score 0.65 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 7 light-curve dip in this sector: 82.6 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 31.51 (gate ≥ 5.0) Compactness 0.85 (gate ≥ 0.50)
Dimming centroid offset 0.07 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 307453353
1.14 px away, dTmag +2.32 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

